

2026 AI前沿英语训练素材

昆明理工大学 计算机专业复试备考
20篇精选英文段落 · 双语翻译 · 60个核心词汇

本材料参考历年昆明理工大学计算机专业复试翻译题目规律，结合2025-2026年人工智能与前沿技术发展动态精心编写。涵盖大型语言模型、多模态AI、AI智能体、量子计算、脑机接口等前沿领域，兼顾英译中与中译英双向练习需求。

第一部分：英文训练段落（20篇）

Passage 1 大型语言模型 (LLMs)

【英文原文】

Large Language Models (LLMs) are a class of foundation models trained on vast amounts of textual data using self-supervised learning techniques. By leveraging transformer architectures with billions of parameters, LLMs acquire remarkable emergent capabilities — including in-context learning, chain-of-thought reasoning, and instruction following — that were not explicitly programmed. Models such as GPT-4, Claude, and Gemini have demonstrated human-level performance across a broad range of benchmarks, from complex mathematics to creative writing, marking a paradigm shift in artificial intelligence research.

【参考译文】

大型语言模型（LLMs）是一类基础模型，通过自监督学习技术在海量文本数据上训练而成。借助拥有数十亿参数的Transformer架构，LLMs获得了显著的涌现能力——包括上下文学习、链式思维推理和指令遵循——这些能力并非被显式编程赋予。GPT-4、Claude和Gemini等模型在从复杂数学到创意写作的广泛基准测试中表现出人类水平，标志着人工智能研究的范式转变。

题源参考：NeurIPS 2024 / OpenAI技术报告

Passage 2 多模态人工智能 (Multimodal AI)

【英文原文】

Multimodal AI systems are designed to process and integrate information from multiple modalities simultaneously, including text, images, audio, and video. Unlike unimodal systems that operate within a single domain, multimodal models such as GPT-4o and Gemini 1.5 Pro can interpret complex cross-modal relationships. This capability enables applications ranging from medical image analysis — where visual findings are correlated with clinical notes — to autonomous driving systems that fuse camera, LiDAR, and sensor data in real time.

【参考译文】

多模态人工智能系统旨在同时处理和整合来自多种模态的信息，包括文本、图像、音频和视频。与在单一领域内运行的单模态系统不同，GPT-4o和Gemini 1.5 Pro等多模态模型能够解读复杂的跨模态关系。这一能力使其得以应用于从医学图像分析（将视觉发现与临床记录相关联）到实时融合摄像头、激光雷达和传感器数据的自动驾驶系统等广泛领域。

题源参考：Google DeepMind技术报告 / CVPR 2024

Passage 3 检索增强生成 (RAG)

【英文原文】

Retrieval-Augmented Generation (RAG) is an architectural paradigm that enhances generative AI models by coupling them with an external knowledge retrieval mechanism. During inference, relevant documents or data chunks are retrieved from a vector database using semantic similarity search, and are then provided as contextual input to the language model. RAG effectively mitigates hallucination — a critical limitation of pure parametric models — by grounding responses in verified, up-to-date information, making it particularly valuable for enterprise knowledge management and question-answering systems.

【参考译文】

检索增强生成（RAG）是一种架构范式，通过将生成式AI模型与外部知识检索机制相结合来增强其能力。在推理阶段，利用语义相似性搜索从向量数据库中检索相关文档或数据块，并作为上下文输入提供给语言模型。RAG通过将回答建立在经过验证的最新信息基础上，有效缓解了幻觉问题——这是纯参数化模型的关键局限——使其在企业知识管理和问答系统中尤具价值。

题源参考：ACL 2024 / Meta AI Research

Passage 4 扩散模型与生成式AI (Diffusion Models)

【英文原文】

Diffusion models have emerged as the dominant paradigm in generative AI for image and video synthesis. These probabilistic models operate through a forward process that gradually adds Gaussian noise to training data, and a learned reverse process that iteratively denoises a random signal to produce high-fidelity outputs. Unlike Generative Adversarial Networks (GANs), diffusion models offer superior training stability and output diversity. The advent of latent diffusion models — exemplified by Stable Diffusion — further reduced computational overhead by operating in a compressed latent space, democratizing high-quality content creation.

【参考译文】

扩散模型已成为图像和视频合成领域生成式AI的主流范式。这些概率模型通过一个前向过程运行，该过程逐渐向训练数据添加高斯噪声，并通过一个学习到的逆向过程迭代去噪随机信号以生成高保真输出。与生成对抗网络（GANs）不同，扩散模型具有更优越的训练稳定性和输出多样性。以Stable Diffusion为代表的潜在扩散模型通过在压缩的潜在空间中运行，进一步降低了计算开销，使高质量内容创作得以普及。

题源参考：ICLR 2024 / Stability AI

Passage 5 AI智能体 (AI Agents)

【英文原文】

AI agents represent a significant evolution beyond conversational AI, characterized by their capacity for autonomous goal-directed behavior within dynamic environments. A modern AI agent perceives its environment through multimodal inputs, maintains a working memory of prior interactions, formulates multi-step plans, and executes actions via tool use — including web browsing, code execution, and API calls. Multi-agent frameworks, where specialized agents collaborate and delegate subtasks, are increasingly applied in software engineering automation, scientific research acceleration, and complex workflow orchestration.

【参考译文】

AI智能体代表了对话式AI的重大演进，其特点是能够在动态环境中进行自主的目标导向行为。现代AI智能体通过多模态输入感知环境，维护先前交互的工作记忆，制定多步骤计划，并通过工具调用（包括网页浏览、代码执行和API调用）执行行动。多智能体框架——专业化智能体相互协作并委派子任务——在软件工程自动化、科学研究加速和复杂 workflow 编排中得到日益广泛的应用。

题源参考：AutoGen / LangGraph 技术文档 2024-2025

Passage 6 联邦学习 (Federated Learning)

【英文原文】

Federated Learning (FL) is a distributed machine learning paradigm in which model training occurs locally on edge devices — such as smartphones or hospital servers — without raw data ever leaving the device. Only model gradient updates are aggregated by a central server, thereby preserving data privacy and complying with regulations such as

GDPR and HIPAA. Recent advances, including differential privacy mechanisms and secure aggregation protocols, have addressed concerns about gradient inversion attacks, making FL increasingly viable for privacy-sensitive applications in healthcare diagnostics, financial fraud detection, and personalized recommendation systems.

【参考译文】

联邦学习（FL）是一种分布式机器学习范式，模型训练在边缘设备（如智能手机或医院服务器）上本地进行，原始数据无需离开设备。只有模型梯度更新由中央服务器聚合，从而保护数据隐私并符合GDPR和HIPAA等法规。包括差分隐私机制和安全聚合协议在内的近期进展已解决了梯度逆向攻击的担忧，使联邦学习在医疗诊断、金融欺诈检测和个性化推荐系统等隐私敏感应用中日益可行。

题源参考：IEEE TPAMI / Google Research 2024

Passage 7 图神经网络 (Graph Neural Networks)

【英文原文】

Graph Neural Networks (GNNs) are a class of deep learning models specifically designed to operate on graph-structured data, where entities are represented as nodes and their relationships as edges. Through iterative message-passing mechanisms, GNNs aggregate neighborhood information to learn rich node, edge, and graph-level representations. They have achieved state-of-the-art results in diverse domains, including molecular property prediction for drug discovery, social network analysis, recommendation systems, and knowledge graph completion. Graph Transformers, which combine attention mechanisms with graph inductive biases, represent the current frontier of the field.

【参考译文】

图神经网络（GNNs）是一类专门用于处理图结构数据的深度学习模型，其中实体表示为节点，关系表示为边。通过迭代消息传递机制，GNNs聚合邻域信息以学习丰富的节点、边和图级别表征。它们在多个领域取得了最先进的成果，包括用于药物发现的分子性质预测、社交网络分析、推荐系统和知识图谱补全。结合注意力机制与图归纳偏置的图Transformer代表了该领域的当前前沿。

题源参考：KDD 2024 / Stanford AI Lab

Passage 8 量子计算与AI (Quantum Computing & AI)

【英文原文】

Quantum computing harnesses the principles of quantum mechanics — superposition, entanglement, and interference — to perform computations that are intractable for classical computers. In the context of artificial intelligence, quantum machine learning algorithms theoretically offer exponential speedups for specific tasks such as optimization problems, sampling from complex probability distributions, and linear algebra operations. While current noisy intermediate-scale quantum (NISQ) devices remain limited by decoherence and gate error rates, hybrid quantum-classical architectures are actively explored as a near-term path toward quantum advantage in AI workloads.

【参考译文】

量子计算利用量子力学原理——叠加态、量子纠缠和量子干涉——来执行经典计算机难以处理的计算。在人工智能领域，量子机器学习算法理论上可为特定任务（如优化问题、从复杂概率分布中采样和线性代数运算）提供指数级加速。尽管当前含噪声的中等规模量子（NISQ）设备仍受限于退相干和门错误率，混合量子-经典架构正被积极探索，作为AI工作负载中实现量子优势的近期路径。

题源参考：Nature / IBM Quantum 2024

Passage 9 神经符号AI (Neuro-Symbolic AI)

【英文原文】

Neuro-symbolic AI is an interdisciplinary research paradigm that seeks to integrate the pattern recognition strengths of neural networks with the logical reasoning capabilities of symbolic AI systems. Pure neural approaches excel at perception and generalization from data but lack explicit reasoning transparency, while symbolic systems offer interpretability and compositionality but struggle with real-world noise and scalability. Hybrid neuro-symbolic architectures, such as those combining large language models with formal logic solvers or knowledge graphs, aim to achieve robust reasoning, systematic generalization, and causality — capabilities essential for the next generation of trustworthy AI.

【参考译文】

神经符号AI是一种跨学科研究范式，旨在将神经网络的模式识别优势与符号AI系统的逻辑推理能力相结合。纯神经网络方法擅长感知和从数据中泛化，但缺乏明确的推理透明度；而符号系统提供可解释性和组合性，却难以应对现实世界的噪声和可扩展性问题。将大型语言模型与形式逻辑求解器或知识图谱相结合的混合神经符号架构，旨在实现鲁棒推理、系统泛化和因果关系——这些能力对于下一代可信AI至关重要。

题源参考：AAAI 2025 / MIT CSAIL

Passage 10 AI安全与对齐 (AI Safety & Alignment)

【英文原文】

AI alignment refers to the technical and governance challenge of ensuring that increasingly capable AI systems reliably pursue goals and values that are beneficial to humanity. Key technical research areas include reinforcement learning from human feedback (RLHF), constitutional AI, interpretability and mechanistic understanding of neural network internals, and scalable oversight methods. As frontier AI models approach and potentially surpass human expert performance across many domains, the field of AI safety has gained critical importance — encompassing not only preventing misuse, but also ensuring robust behavior under distributional shift and averting catastrophic risks from misaligned superintelligent systems.

【参考译文】

AI对齐是指确保能力日益增强的AI系统可靠地追求对人类有益的目标和价值观这一技术与治理挑战。关键技术研究领域包括基于人类反馈的强化学习（RLHF）、宪法AI、神经网络内部的可解释性与机制理解，以及可扩展监督方法。随着前沿AI模型在许多领域接近乃至超越人类专家水平，AI安全领域已获得至关重要的地位——不仅涵盖防止滥用，还包括确保在分布偏移下的鲁棒行为，以及规避来自未对齐超级智能系统的灾难性风险。

题源参考：Anthropic / OpenAI Safety Report 2024

Passage 11 边缘AI与物联网 (Edge AI & IoT)

【英文原文】

Edge AI refers to the deployment of artificial intelligence models directly on edge devices — such as smartphones, embedded systems, autonomous vehicles, and industrial IoT sensors — rather than relying on cloud-based inference. This paradigm offers critical advantages including ultra-low latency, reduced bandwidth consumption, enhanced data privacy, and offline operational capability. Key enabling technologies include model compression techniques (quantization, pruning, knowledge distillation), neuromorphic chips, and dedicated neural processing units (NPUs). The convergence of 5G connectivity with edge AI is accelerating the realization of intelligent

manufacturing, smart cities, and precision agriculture.

【参考译文】

边缘AI是指直接在边缘设备上部署人工智能模型——如智能手机、嵌入式系统、自动驾驶汽车和工业物联网传感器——而非依赖基于云的推理。这一范式提供了关键优势，包括超低延迟、降低带宽消耗、增强数据隐私和离线操作能力。关键使能技术包括模型压缩技术（量化、剪枝、知识蒸馏）、神经形态芯片和专用神经处理单元（NPUs）。5G连接与边缘AI的融合正在加速智能制造、智慧城市和精准农业的实现。

题源参考：IEEE IoT Journal / Qualcomm AI Research 2024

Passage 12 自然语言处理前沿 (NLP Frontiers)

【英文原文】

The field of Natural Language Processing has undergone a profound transformation driven by the scaling law hypothesis, which posits that model performance improves predictably as a power function of compute, data, and parameters. The emergence of in-context learning — where a model adapts to a new task solely through demonstration examples in the prompt, without any gradient updates — has fundamentally challenged the traditional fine-tuning paradigm. Furthermore, chain-of-thought prompting has unlocked complex multi-step reasoning capabilities in LLMs, while techniques such as tool use and function calling have expanded their operational scope from text generation to dynamic real-world interaction.

【参考译文】

自然语言处理领域经历了由缩放定律假说驱动的深刻变革，该假说认为模型性能随着计算量、数据量和参数量的幂函数而可预测地提升。上下文学习的涌现——模型仅通过提示中的示例就能适应新任务，无需任何梯度更新——从根本上挑战了传统的微调范式。此外，链式思维提示解锁了LLMs的复杂多步推理能力，而工具调用和函数调用等技术将其操作范围从文本生成扩展到了动态现实世界交互。

题源参考：ACL 2024 Anthology / DeepMind

Passage 13 计算机视觉最新进展 (Computer Vision Advances)

【英文原文】

The Vision Transformer (ViT) architecture has fundamentally reshaped the landscape of computer vision by applying the self-attention mechanism — originally developed for NLP — directly to sequences of image patches. This has enabled models to capture global contextual relationships across an entire image, overcoming the locality bias inherent in convolutional neural networks. Subsequent innovations, including the Segment Anything Model (SAM) and DINOv2, have demonstrated zero-shot and few-shot generalization capabilities that approach human-level performance on complex visual understanding tasks such as panoptic segmentation, 3D scene reconstruction, and video object tracking.

【参考译文】

视觉Transformer (ViT) 架构通过将最初为NLP开发的自注意力机制直接应用于图像块序列，从根本上重塑了计算机视觉的格局。这使模型能够捕获整张图像的全局上下文关系，克服了卷积神经网络固有的局部性偏置。后续创新，包括Segment Anything Model (SAM) 和DINOv2，展示了零样本和少样本泛化能力，在全景分割、3D场景重建和视频目标跟踪等复杂视觉理解任务上接近人类水平。

题源参考：ECCV 2024 / Meta AI

Passage 14 强化学习与决策 (Reinforcement Learning)

【英文原文】

Reinforcement Learning (RL) is a machine learning paradigm in which an agent learns optimal decision-making policies through trial-and-error interactions with an environment, receiving scalar reward signals that guide its behavior. The integration of deep neural networks with RL — known as Deep Reinforcement Learning (DRL) — has yielded remarkable breakthroughs, from AlphaGo defeating world champion Go players to AlphaFold revolutionizing protein structure prediction. More recently, Reinforcement Learning from Human Feedback (RLHF) has become the cornerstone technique for aligning large language models with human preferences, enabling the development of helpful, harmless, and honest AI assistants.

【参考译文】

强化学习（RL）是一种机器学习范式，其中智能体通过与环境的试错交互来学习最优决策策略，接收指导其行为的标量奖励信号。深度神经网络与RL的结合——称为深度强化学习（DRL）——取得了显著突破，从AlphaGo击败围棋世界冠军到AlphaFold革新蛋白质结构预测。更近期，基于人类反馈的强化学习（RLHF）已成为将大型语言模型与人类偏好对齐的核心技术，推动了有益、无害且诚实的AI助手的开发。

题源参考：DeepMind / Anthropic Constitutional AI 2024

Passage 15 AI芯片与硬件加速 (AI Chips & Hardware)

【英文原文】

The rapid advancement of artificial intelligence has been critically enabled by innovations in specialized hardware architectures. Graphics Processing Units (GPUs), originally designed for rendering, proved serendipitously well-suited for the massively parallel matrix operations that underpin deep learning. The development of dedicated AI accelerators — including Google's Tensor Processing Units (TPUs), Apple's Neural Engines, and NVIDIA's H100 and Blackwell series — has delivered orders-of-magnitude improvements in training throughput and inference efficiency. In-memory computing architectures and photonic computing represent promising frontiers that aim to overcome the von Neumann bottleneck in AI workloads.

【参考译文】

人工智能的快速发展在很大程度上得益于专用硬件架构的创新。图形处理单元（GPU）最初为渲染设计，却意外地非常适合支撑深度学习的大规模并行矩阵运算。专用AI加速器的开发——包括谷歌的张量处理单元（TPUs）、苹果的神经引擎以及英伟达的H100和Blackwell系列——在训练吞吐量和推理效率方面带来了数量级的提升。存算一体架构和光子计算代表了有望克服AI工作负载中冯·诺依曼瓶颈的前沿方向。

题源参考：Hot Chips 2024 / NVIDIA GTC

Passage 16 生物信息学与AI (Bioinformatics & AI)

【英文原文】

The application of artificial intelligence to bioinformatics has catalyzed unprecedented breakthroughs in life sciences. AlphaFold 3, developed by Google DeepMind, extended its groundbreaking protein structure prediction capabilities to model the interactions between proteins, DNA, RNA, and small molecules with near-experimental accuracy, dramatically accelerating drug discovery pipelines. Large language models pre-trained on genomic sequences — termed genomic foundation models — have demonstrated remarkable performance in predicting gene expression, identifying disease-associated variants, and designing novel therapeutic proteins. This convergence of AI and biology is increasingly regarded as a defining scientific frontier of the 21st century.

【参考译文】

人工智能在生物信息学中的应用催生了生命科学领域前所未有的突破。由Google DeepMind开发的AlphaFold 3将其开创性的蛋白质结构预测能力扩展到以接近实验精度模拟蛋白质、DNA、RNA和小分子之间的相互作用，显著加速了药物发现流程。在基因组序列上预训练的大型语言模型——称为基因组基础模型——在预测基因表达、识别疾病相关变异和设计新型治疗性蛋白质方面表现出卓越性能。AI与生物学的这种融合日益被视为21世纪决定性的科学前沿。

题源参考：Nature / DeepMind AlphaFold 3 论文 2024

Passage 17 负责的AI与伦理 (Responsible AI & Ethics)

【英文原文】

Responsible AI encompasses a multidisciplinary framework for ensuring that AI systems are developed and deployed in a manner that is fair, transparent, accountable, and aligned with human values. Algorithmic bias — where models systematically produce discriminatory outputs due to imbalances in training data or flawed objective functions — poses significant risks to social equity in high-stakes domains such as hiring, credit scoring, and criminal justice. Explainable AI (XAI) techniques, including LIME, SHAP, and mechanistic interpretability research, aim to provide human-understandable insights into model decision processes, enabling meaningful human oversight and regulatory compliance.

【参考译文】

负责的AI涵盖了一个多学科框架，旨在确保AI系统的开发和部署方式公平、透明、可问责，并与人类价值观相一致。算法偏见——模型由于训练数据不平衡或目标函数缺陷而系统性地产生歧视性输出——在招聘、信用评分和刑事司法等高风险领域对社会公平构成重大风险。可解释AI（XAI）技术，包括LIME、SHAP和机制可解释性研究，旨在提供人类可理解的模型决策过程洞察，实现有意义的人类监督和监管合规。

题源参考：EU AI Act 2024 / FAccT Conference

Passage 18 自动驾驶技术 (Autonomous Driving)

【英文原文】

Autonomous driving systems integrate a sophisticated perception-prediction-planning pipeline to enable vehicles to navigate complex real-world environments without human intervention. Modern end-to-end learning approaches, exemplified by Tesla's FSD and Waymo's architecture, leverage transformer-based models to process rich multimodal sensor fusion from cameras, LiDAR, and radar, producing driving actions directly from raw sensor inputs. A central challenge remains the long-tail distribution of rare but critical edge cases — unusual weather conditions, ambiguous traffic scenarios, and unpredictable human behavior — that demand robust generalization capabilities far beyond what current training datasets can adequately cover.

【参考译文】

自动驾驶系统整合了复杂的感知-预测-规划流程，使车辆能够在无需人工干预的情况下在复杂的现实环境中导航。现代端到端学习方法，以特斯拉的FSD和Waymo的架构为代表，利用基于Transformer的模型处理来自摄像头、激光雷达和雷达的丰富多模态传感器融合，直接从原始传感器输入产生驾驶动作。核心挑战仍然是罕见但关键的边缘案例的长尾分布——异常天气条件、模糊的交通场景和不可预测的人类行为——这需要远超当前训练数据集所能充分覆盖的鲁棒泛化能力。

题源参考：IEEE IV 2024 / Waymo Safety Report

Passage 19 AI代码生成 (AI Code Generation)

【英文原文】

AI-powered code generation has matured from simple autocomplete tools into sophisticated systems capable of implementing entire software modules from natural language specifications. Models such as GitHub Copilot, powered by OpenAI Codex, and Anthropic's Claude have demonstrated proficiency in generating syntactically correct and functionally coherent code across dozens of programming languages. The emergence of AI software engineering agents — systems that can autonomously decompose complex tasks, write and execute code, interpret error messages, and iteratively refine solutions — signals a fundamental transformation in software development workflows, with implications for developer productivity, software quality assurance, and the evolving role of human engineers.

【参考译文】

AI驱动的代码生成已从简单的自动补全工具发展为能够根据自然语言规范实现整个软件模块的复杂系统。由OpenAI Codex驱动的GitHub Copilot和Anthropic的Claude等模型已展示出在数十种编程语言中生成语法正确且功能连贯代码的能力。AI软件工程智能体的出现——能够自主分解复杂任务、编写和执行代码、解读错误信息并迭代优化解决方案的系统——标志着软件开发工作流程的根本性变革，对开发人员生产力、软件质量保证和人类工程师的演变角色具有深远影响。

题源参考：GitHub Copilot / SWE-bench 2024

Passage 20 脑机接口与神经技术 (BCI & Neurotechnology)

【英文原文】

Brain-Computer Interfaces (BCIs) establish direct communication pathways between the biological neural activity of the brain and external computational or physical devices, bypassing conventional neuromuscular channels. Non-invasive BCIs utilizing electroencephalography (EEG) have enabled paralyzed patients to control prosthetic limbs and communicate through thought alone. Neuralink's implantable N1 chip, with its high-density microelectrode arrays capable of recording from thousands of neurons simultaneously, represents the state of the art in invasive BCIs. The integration of AI with BCIs — particularly for real-time neural signal decoding and adaptive stimulation — is opening new frontiers in neurological rehabilitation, augmented cognition, and the long-term vision of seamless human-AI symbiosis.

【参考译文】

脑机接口（BCIs）在大脑的生物神经活动与外部计算或物理设备之间建立直接通信路径，绕过传统的神经肌肉通道。利用脑电图（EEG）的非侵入式BCI已使瘫痪患者能够控制假肢并仅凭思维进行交流。Neuralink的可植入N1芯片，凭借其能够同时从数千个神经元记录的高密度微电极阵列，代表了侵入式BCI的最先进水平。AI与BCI的整合——特别是在实时神经信号解码和自适应刺激方面——正在为神经康复、增强认知以及人机无缝共生的长远愿景开辟新的前沿。

题源参考：Neuralink / Nature Neuroscience 2024

第二部分：AI前沿核心词汇（60个）

英文术语	中文译名	释义
transformer	Transformer架构	深度学习中的注意力机制架构
tokenization	分词/标记化	将文本拆分为token的过程
embedding	嵌入	将对象映射为向量的表示
inference	推理/推断	模型利用已有参数进行预测
fine-tuning	微调	在预训练基础上针对特定任务训练
hallucination	幻觉	AI生成不准确或虚假内容的现象
prompt engineering	提示工程	设计输入以优化模型输出的技术
hyperparameter	超参数	模型训练前人工设定的参数
gradient descent	梯度下降	通过计算梯度更新模型参数的优化算法
backpropagation	反向传播	计算神经网络梯度的核心算法
overfitting	过拟合	模型过度拟合训练数据，泛化能力下降
regularization	正则化	防止过拟合的技术手段
batch normalization	批归一化	对每层输入进行归一化以加速训练
attention mechanism	注意力机制	允许模型聚焦于输入中相关部分的机制
self-attention	自注意力	序列元素之间相互关注的注意力机制
autoregressive	自回归	逐步生成序列，每步依赖前序输出
latent space	潜在空间	数据的高维压缩表示空间
zero-shot learning	零样本学习	无需标注样本即可完成新任务
few-shot learning	少样本学习	仅用极少样本完成任务学习
knowledge distillation	知识蒸馏	将大模型知识迁移到小模型的技术
quantization	量化	降低模型权重精度以减少计算开销
pruning	剪枝	移除冗余神经元或连接以压缩模型
scalability	可扩展性	系统处理不断增长负载的能力
benchmark	基准测试	评估模型性能的标准测试集
generalization	泛化	模型在未见数据上的表现能力
foundation model	基础模型	在大规模数据上预训练的通用模型
emergent capability	涌现能力	模型规模增大时出现的意外新能力
chain-of-thought	链式思维	引导模型逐步推理的提示策略
RLHF	基于人类反馈的强化学习	利用人类偏好数据对齐AI行为

英文术语	中文译名	释义
vector database	向量数据库	存储和检索高维向量的专用数据库
semantic search	语义搜索	基于含义而非关键词的搜索技术
multimodal	多模态	处理多种数据类型（文本/图像/音频等）
neural architecture search	神经架构搜索	自动化搜索最优网络结构的技术
convolutional layer	卷积层	提取局部特征的神经网络层
pooling layer	池化层	降采样以减少特征图尺寸的层
dropout	随机失活	训练时随机丢弃神经元防过拟合
residual connection	残差连接	跳跃连接缓解深层网络梯度消失
encoder-decoder	编码器-解码器	将输入编码再解码输出的架构
generative model	生成模型	能够生成新数据样本的模型
discriminative model	判别模型	学习类别边界的分类模型
reinforcement learning	强化学习	通过奖惩信号学习最优策略
reward function	奖励函数	强化学习中量化行为好坏的函数
policy gradient	策略梯度	直接优化策略参数的强化学习算法
explainability	可解释性	AI决策过程可被人类理解的属性
bias-variance tradeoff	偏差-方差权衡	模型复杂度与泛化误差的平衡
data augmentation	数据增强	通过变换扩充训练数据的技术
transfer learning	迁移学习	将一个任务上的知识迁移到新任务
federated learning	联邦学习	保护隐私的分布式机器学习范式
differential privacy	差分隐私	在统计输出中保护个体数据隐私的机制
neural processing unit	神经处理单元	专为AI推理设计的芯片
diffusion model	扩散模型	通过去噪过程生成数据的生成模型
GAN	生成对抗网络	生成器与判别器博弈的生成模型框架
contrastive learning	对比学习	通过拉近正样本、推远负样本的自监督方法
graph neural network	图神经网络	处理图结构数据的神经网络
knowledge graph	知识图谱	以图结构表示实体及其关系的知识库
in-context learning	上下文学习	仅用提示示例而无梯度更新完成任务
model alignment	模型对齐	使AI目标与人类价值观保持一致
hallucination mitigation	幻觉缓解	减少AI生成虚假内容的技术手段
autonomous agent	自主智能体	能自主感知、规划并行动的AI系统
edge computing	边缘计算	在靠近数据源的设备上进行计算
neuromorphic computing	神经形态计算	模仿生物神经系统的计算架构

